

Course: MS&E 152 summer 2026
Sequence: Week 1, Lecture 1
Date: Monday, June 22nd 2026
Topic: Introduction to Decision Analysis

Links:

Course website: <https://stanford-msande152.github.io/summer26/>
Canvas: <https://canvas.stanford.edu/courses/228284>

Title: Ethical Analysis

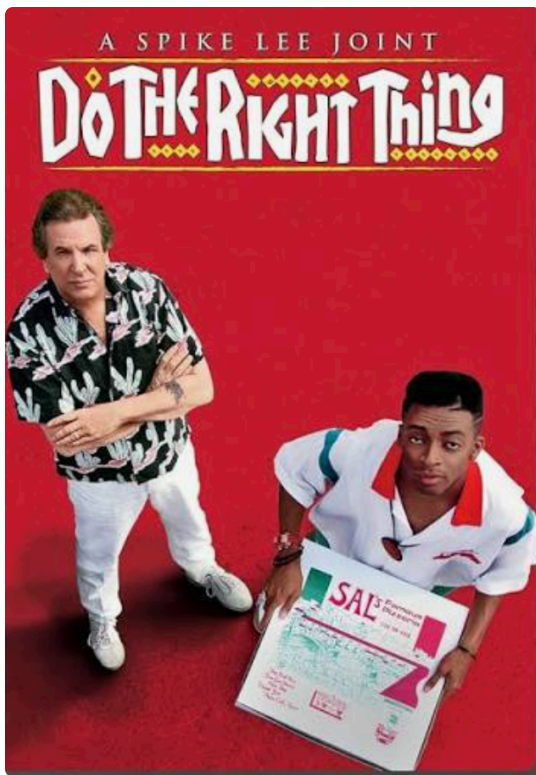
What you will learn

Ethical questions that arise in Decision Analysis

Class schedule

- What is Ethics, ethics in professional practice
 - Decision Analysis and ethics
 - Winding up the class.
 - Class presentation session.
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What is Ethics?



(Movie, ["Do the Right Thing"](#), Spike Lee, 1989)

The question: Doing the *right* thing. Is what we are doing *good*?

We started off this class with the question "What is a *good* decision." Our answer has been to apply the five tenets of actional thought.

What is a consequence of the tenets we started with? Is there "rationality" that is not just *individual* rationality? What about a choice that might lead to harm to others? Most glaringly because in our work we are just one member of a group, within society, with different values, we need to consider the social aspect of our action.

The question "What is good" opens the door to a question people have pondered for some time:

It is sometimes said, either irritably or with some satisfaction, that philosophy makes no progress. ... this is an abiding and not a regrettable characteristic of the discipline, that philosophy has... to keep trying to return to the beginning. (Murdoch, p. 1)

What we will cover in this class

Our goal, as practicing engineers (Decision Analysts) is how do we devise a method that we can apply in our professional practice? One should think about it, so that you have the ethical skills so that when situations arise you have your framework to fall back on. We treat the terms "moral" and "ethical" as synonyms

In this lecture we can only define the question -- what is ethics. There's a full course on "The Ethical Analyst" that follows this course.

A point of comparison - Classical Virtue Ethics - Stoicism

Stoicism is a branch of Greco-Roman philosophy that sees ethics as an aspect of personal virtue. Its notion of a life well lived is to live in agreement with our nature. Particular to our human nature is "the ability to reason and the fact that we are highly social." (Pigliucci & Lopez, p. 84) From these follow four principal virtues: Wisdom, Temperance, Courage and Justice, . Let's interpret these in turn, in current Decision Analysis terms, to see how it matches up.

What is the ethical content of DA?

Lets start with some ethical concepts (the Stoic's virtues) -- how one should act in one's life (a broader concept than rules of right and wrong.)

For sake of argument, consider how DA stacks up to these four "cardinal virtues".

- **Wisdom.** (Reason) The term encompasses appeal to reason from principles. Thinking through the aspects of how to act, including longer term and indirect consequences. DA offers a principled basis for reasoning So the derivation of decision theory and its application from basic tenets would consider Decision Analysis to be wise, but wisdom may draw upon a wider basis of first principles.
- **Temperance.** (Moderation) Matching one's actions to the situation at hand. Here Decision Analysis is a major advance over classical thinking, specifically Aristotle's notion of that an ethical action sits at the mean between extremes. He gives an example from medicine and health. To paraphrase -- what is appropriate is destroyed by either deficiency or excess as in the case exercise or nutrition, however proportionate amounts create and preserve health. (my version Nichomachean Ethics Book 2, Ch 3, 1104a15.) The ancients did not have the benefit of quantifiable probability that was only developed relatively recently, by which Aristotle's idea of "proportionate" action is made precise. However temperance implies more than just the "prudential" pursuit of ends via means.
- **Courage** (commitment)The term is better understood nowadays as "commitment" or "orientation to action." Acting on what one determines is good. Having the will to realize what is good. DA removes regret via "clarity of action", so to remove any restraints on making the best choice. Without a willingness to act all other aspects are vacuous.
- **Justice** (fairness) Giving each person what is due them. Do no harm to others. A person whose wisdom I respect said that the first question to ask of any action is *whom does it affect?* and on this criterion Decision Analysis is mute. In this area we need a way to advance beyond individual economic rationality as derived from the five tenets. Individual rationality was never intended as a solution for group decision making.

In short, our notion of a good decision partly matches what a Stoic philosopher regards as ethical. Decision analysis can be called "prudent", it guides one in being prudent. In Stoic thought any action has to respect all virtues to be ethical. Decision analysis contributes the prudent part. However since the Stoic requires a good action to incorporate all virtues, to be ethical we need to extend the idea of a Good Decision, specifically to consider how an action affects other people.

Class Activity

Please take the class evaluation survey

Class presentations here, tomorrow from 2 to 4 PM. [Sign up on this spreadsheet.](#)

Presenters - send me your slides (powerpoint or pdf) so we can show them from my machine.

Want to keep in touch? [Connect to me on LinkedIn.](#)

Key terms

Good

Ethical, moral

Stoic Virtue: Wisdom, Temperance, Courage, Justice

Prudence

Files, references

- R.C. Bartlett, S.D. Collins (2011) translators, "Aristotle's Nicomachean Ethics" , U. of Chicago Press.
- R. Howard, C. Korver (2008) "Ethics for the Real World" Harvard Business Press, MA.
- I. Murdoch (1970) "The Sovereignty of Good" Routledge Classics, NY.
- M. Pigliucci, G. Lopez (2025)"Beyond Stoicism" The Experiment LLC, NY.

Curious? Things to explore

Stanford Encyclopedia of Philosophy

<https://altiolem.org/organisation/stanford-encyclopedia-of-philosophy>

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